Kota Yasasvi Surya Vignesh

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EDUCATION

University of Petroleum and Energy Studies, India — **July 2023 – May 2027**
Bachelor of Technology in Aerospace Engineering — CGPA: 7.05/10.0

Sri Chaitanya Junior College, Andhra Pradesh — **March 2023**
Board of Intermediate Andhra Pradesh — Percentage: 85.3%

Sri Chaitanya Techno School, Andhra Pradesh — **May 2021**
State Board of Andhra Pradesh — Percentage: 97.6%

SKILLS

Software: AutoCAD, SolidWorks, CATIA V5, OpenRocket, XFLR5, COMSOL, ANSYS, Systems Tool Kit (STK).
Programming: Python, MATLAB.
Soft Skills: Leadership, Adaptability, Time management.

INTERNSHIP

Prime Toolings — **June 2026 - July 2026**
Intern — **Bangalore**
- Engineered RDE/Ramjet combustors, aerospike nozzles, and turbomachinery in CATIA, running reactive CFD in ANSYS.
- Executed system coupling for RDE test rigs and analyzed supersonic aerodynamics and combustion on missile geometries.

Hypstuma — **June 2025 - July 2025**
Technical Intern — **Chennai**
- Engineered Mechanical UAV Components Using Solid Works, Focusing on Structural Integrity and Design Optimization
- Worked as Team Lead for a Research on Performance and Analysis of DJI Matrice 350 RTK Drone.

India Space Lab — **June 2025 - July 2025**
Trainee — **Remote**
- Studied the fundamental architecture and integration of small-scale satellite systems
- Analyzed rocketry designs through OpenRocket to optimize stability, propulsion, and recovery system deployment.

Souls Of Humanity (NGO) — **June 2024 - July 2024**
Team Lead — **Visakhapatnam**
- Assisted in the Mission Vidya Vikas project, facilitating educational programs for underprivileged children.
- Worked as Team Lead at Child Development Centre and worked for Children's Education and their Awareness of Society.

PROJECTS

Characterization of Primary Atomization in Liquid Rocket Engines | Major Project — **Aug 2026 - Present**
- Conducting hydrodynamic calculations & multiphase CFD to characterize injector spray breakup & droplet dynamics.
- Executing experimental cold-flow spray tests to measure droplet size distribution and optimize propellant mixing efficiency.

Differential Drive Robot for Aircraft Inspection | Minor Project — **Aug 2025 - May 2026**
- Engineered a sensor-integrated platform using ESP32 & 30 RPM motors to inspect aircraft fuselage panels and runway surface.
- Achieved a 90% defect detection accuracy for cracks as small as 0.2 mm.

Remaining Useful Lifetime Estimation For Gas Turbine Engines | Capstone Project — **Jan 2026 - May 2026**
- Developed a predictive maintenance pipeline using Python to estimate the RUL using the NASA C-MAPSS datasets.
- Evaluated multiple Models including Random Forest, LSTM, and Bi-LSTM, achieving a RMSE of 29 with the LSTM model.

Performance Analysis & Flight Envelope of Dassault Rafale | PBL Project — **Apr 2026 - June 2026**
- Evaluating the aerodynamic performance of Rafale by analytical calculations and XFLR5 simulation using the NACA 23012.
- Simulating a scaled-down CAD model in ANSYS Fluent to analyze lift-drag characteristics and validate the V-n diagram.

Structural Analysis of Tapered Fighter Aircraft Wing | Structures Project — **Apr 2026 - May 2026**
- Designing a tapered fighter wing using the NACA 23012, with a structural framework of 2 spars and transverse ribs.
- Evaluating structural endurance in ANSYS by applying pressure loads to validate analytical results against simulation data.

Comparative CFD Investigation of Aerospike and CD Nozzles at Different Altitudes | CFD Project — **July 2026 - Present**
- Conducting a comparative CFD study in ANSYS evaluating Aerospike vs. CD nozzle performance across varying altitude.
- Analyzing thrust efficiency, mass flow rates, and contours to evaluate pressure-thrust compensation from sea level to 15 km.

Drone-based Thermal inspection of Photovoltaic Panels for Error/fault Detection | CAD Project — **Feb 2026 - Present**
- Designed a custom CAD model of a UAV integrated with a thermal imaging camera for automated solar farm inspection.
- Executing aerodynamic & structural to optimize flight stability and enable precise, GPS-mapped hotspot and defect detection.

ACCOMPLISHMENTS

- Certificate for Computational Fluid Dynamics using Ansys by Udemy.
- Certified in MATLAB Onramp – MathWorks.
- Certified in SolidWorks (Basic to Advanced**)** – Eduonix.
- Certified in Basics of CATIA V5 – Skill-Lync (2026).
- Presented Poster on "Sustainable Aviation" in Energy Summit 2024.